

NIRMA UNIVERSITY
School of Technology, Institute of Technology
Master of Computer Application (2-Years Programme)
Semester-I

6CS104-Database Management System

List of Practical

3	Write and Perform the SQL queries for the following: 1. From the following table, create a view for those salespeople who belong to the city of New York. CREATE or REPLACE VIEW sp AS SELECT * FROM salespeople WHERE city = 'Surat'; table: salespeople select * from sp; 2. From the following table, create a view for all salespeople. Return salesperson ID, name, and city. CREATE or REPLACE VIEW view_sp1 AS SELECT snum, sname, city FROM salespeople; table: salespeople select * from view_sp1; 3. From the following table, create a view to locate the salespeople in the city 'New York'. CREATE VIEW view_sp2 AS SELECT snum, sname, city FROM salespeople WHERE city = 'New York'; table: salespeople select * from view_sp2; 4. From the following table, create a view that counts the number of customers in each city. CREATE VIEW view_sp3 AS SELECT city, COUNT(*) AS cnum FROM customer GROUP BY city; table: customer select * from view_sp3; 5. From the following table, create a view to count the number of unique customers, compute the average and the total purchase amount of customer orders by each date. CREATE VIEW view_sp4 AS SELECT odate, COUNT(DISTINCT cnum) AS unique_customers, AVG(amt) AS avg_purchase_amount, SUM(amt) AS total_purchase_amount FROM orders GROUP BY odate; table : orders select * from view_sp4; 6. From the following tables, create a view to get the salesperson and customer by name. Return order name, purchase amount, salesperson ID, name, customer name. CREATE VIEW view_sp5 AS SELECT o.onum, o.amt, s.snum, s.sname AS salesperson_name, c.cname AS customer_name	1	10	4
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FROM orders o JOIN salespeople s ON o.snum = s.snum JOIN customer c ON o.cnum = c.cnum;

table: salespeople, customer, orders

select * from view_sp5;

7. From the following table, create a view to find the salesperson who handles a customer who makes the highest order of the day. Return order date, salesperson ID, name.

CREATE VIEW sp7 AS

SELECT

o.odate,
s.snum,
s.sname

FROM

orders o

JOIN

salespeople s ON o.snum = s.snum

WHERE

o.amt = (
SELECT MAX(o2.amt)
FROM orders o2
WHERE o2.odate = o.odate
);

table: salespeople, orders

select * from sp7;

8. From the following table, create a view to find the salesperson who deals with the customer with the highest order at least three times per day. Return salesperson ID and name.

CREATE VIEW sp8 AS

SELECT

s.snum,
s.sname

FROM

salespeople s

JOIN

orders o ON s.snum = o.snum

JOIN

customer c ON o.cnum = c.cnum

WHERE

o.amt = (
SELECT MAX(o2.amt)
FROM orders o2
WHERE o2.odate = o.odate
)

GROUP BY

s.snum, s.sname

HAVING

COUNT(o.onum) >= 3;

table: customer, salespeople, orders

select * from sp8;

9. From the following table, create a view to find all the customers who have the highest grade. Return all the fields of customer.

CREATE VIEW sp9 AS

```

SELECT *
FROM customer
WHERE grade = (SELECT MAX(rating) FROM customer);
table: customer
select * from sp9;
10. From the following table, create a view to count the number of
salespeople in each city. Return city, number of salespersons.
CREATE VIEW salespeople_count_by_city AS
SELECT city, COUNT(*) AS number_of_salespersons
FROM salespeople
GROUP BY city;
table: salespeople
select * from salespeople_count_by_city;
11. From the following table, create a view to compute the average purchase
amount and total purchase amount for each salesperson. Return name,
average purchase and total purchase amount. (Assume all names are
unique.).
CREATE VIEW salesperson_purchase_summary AS
SELECT s.sname,
       AVG(o.amt) AS average_purchase_amount,
       SUM(o.amt) AS total_purchase_amount
FROM salespeople s
JOIN orders o ON s.snum = o.snum
GROUP BY s.sname;
Sample table: salespeople, orders
select * from salesperson_purchase_summary;
12. From the following table, create a view to identify salespeople who work
with multiple clients. Return all the fields of salesperson.
CREATE VIEW Salespeople_Multiple_Clients AS
SELECT s.snum, s.sname
FROM salespeople s JOIN customer c ON s.snum = c.snum GROUP BY s.snum, s.sname, s.city
HAVING COUNT(DISTINCT c.cnum) > 1;
Sample table: salespeople, customer
select * from Salespeople_Multiple_Clients;
13. From the following table, create a view that shows all matching
customers with salespeople, ensuring that at least one customer in the city of
the customer is served by the salesperson in the city of the salesperson.
CREATE VIEW sp13 AS
SELECT c.cnum, s.snum
FROM salespeople s
JOIN customer c ON s.snum = c.snum
WHERE s.city = c.city;
Sample table: salespeople, customer
Select * from sp13;
14. From the following table, create a view to display the number of orders
per day. Return order date and number of orders.
CREATE VIEW Orders_Per_Day AS
SELECT odate, COUNT(onum) AS number_of_orders
FROM orders
GROUP BY odate;
Sample table: orders
Select * from Orders_Per_Day;

```

15. From the following table, create a view to find the salespeople who placed orders on 24-Jan-23. Return all the fields of salesperson.

CREATE or replace **VIEW** Salespeople_Orders **AS**

```
SELECT s.snum, s.sname
FROM salespeople s
JOIN orders o ON s.snum = o.snum
WHERE o.odate = '24-Jan-23';
```

Sample table: salespeople, orders

Select * from Salespeople_Orders;

16. From the following table, create a view to find the salespersons who issued orders on either August 17th, 2012 or October 10th, 2012. Return salesperson ID, order number and customer ID.

CREATE VIEW sp16 **AS**

```
SELECT snum, onum, cnum
FROM orders
WHERE odate IN (TO_DATE('2012-08-17', 'YYYY-MM-DD'), TO_DATE('2012-10-10', 'YYYY-MM-DD'));
```

Sample table: orders

select * from sp16;